

TestCaselId	Test Level	Component	Priority
TC-Unit-01	Unit	Customer Authentication	P0
TC-Unit-02	Unit	Administrator Authentication	P0
TC-Unit-03	Unit	Ticket	P0
TC-Unit-04	Unit	Transaction	P0
TC-Integration-01	Integration	Payment_Processing&Transaction_History	P0
TC-Integration-02	Integration	Refund_Processing&Ticket_Status	P0
TC-Integration-03	Integration	Payment_Processing&Ticket_Generation	P0
TC-Integration-04	Integration	Ticket_Generation&Seat_Availability_Update	P0
TC-Functional-01	Functional	Movie_Search	P0
TC-Functional-02	Functional	Submit_Feedback	P0

TC-Functional-03	Functional	Guest_Checkout_Email_Validation	P0
TC-Functional-04	Functional	Guest_Price_Calculation	P0
TC-System-01	System	Customer_Ticket_Purchase_System	P2
TC-System-02	System	Administrator_Refund_Processing_Module	P0

TC-System-03	System	Administrator_Showtime_Management_ Module	P0
TC-System-04	System	Administrator_Movie_Management_ _Module	P0

Description/Test Summary
Verify that the customer authentication component accepts a registered email address and correct password.
Verify that the administrator authentication component rejects an invalid multi-factor authentication code.
Verify that the ticket-generation component creates a unique ticket identifier containing the correct movie, showtime, and seat information.
Verify that the transaction validation component rejects a ticket quantity greater than 20.
Verify that an approved payment creates a completed transaction in the registered customer's purchase history.
Verify that a successful administrator refund changes the associated ticket status to Refunded and restores seat availability.
Verify that the ticket-generation component creates tickets only after the payment service approves the transaction.
Verify that ticket generation after a successful purchase changes the purchased seat from available to sold.
Verify that the movie search bar returns matching movies when a valid movie title is entered.
Verify that feedback cannot be submitted twice by pressing the Submit button in quick succession

Verify that the email address entered during guest ticket purchase is validated before the purchase is completed.

Verify that the total ticket price is calculated correctly, including applicable taxes, a 5% online service fee, and a \$2.99 seat reservation fee, for a regular guest ticket purchase.

Verify that a registered customer can log in, search/browse for an available movie, select a showtime and available seat, complete payment and receive a valid Ticket.

Verify that an administrator can process a refund and that the refunded ticket and seat are updated correctly.

Verify that an administrator can modify an existing showtime and the updated showtime information is saved and displayed throughout the system

Verify that an administrator can add a new movie and new movie is displayed throughout the system correctly.

Pre-requisites
1. The Theater Ticketing System is available. 2. A registered customer account exists with email customer@test.com. 3. The correct password is ValidPass123!.
1. The Theater Ticketing System is available. 2. A valid administrator account exists. 3. The username and password have already been validated. 4. The valid MFA code is 246810.
1. A successful payment result exists. 2. Movie, showtime, and seat information are available. 3. Seat A10 is available.
1. The transaction component is available. 2. A valid movie and showtime are selected. 3. At least 21 seats exist in the test data.
1. A registered customer is logged in. 2. A movie, showtime, and available seat are selected. 3. The payment service test environment is available. 4. Valid test payment information is available.
1. An administrator is authenticated. 2. A completed account-based purchase exists. 3. The showtime has not started. 4. The ticket has not previously been refunded.
1. A valid checkout request exists. 2. The selected seat is available. 3. The payment and ticket-generation services are available.
1. Seat is available for the selected showtime. 2. Payment has been approved. 3. The ticket-generation and seat-management components are available.
1. The Theater Ticketing System is available. 2. The movie "Interstellar" exists in the testing database. 3. A web browser is launched.
1. The Theater Ticketing System is available. 2. A web browser is launched. 3. A ticket purchase process has been completely successfully. 4. The feedback submission form is displayed.

1. The Theater Ticketing System is available.
2. At least one movie, one showtime, and one seat are available.
3. A web browser is launched.

1. The Theater Ticketing System is available.
2. At least one movie, one showtime, and one seat are available.
3. A web browser is launched.
4. The user is not logged into a customer account.

1. The Theater Ticketing System is available.
2. A registered customer account exists.
3. At least one movie, and showtime available.
4. At least one seat is available for selected showtime.
5. The customer has valid payment information to process payment.

1. A valid administrator account exists.
2. A completed ticket purchase with a valid transaction ID exists.
3. The ticket has not already been refunded.

1. The web browser is launched.
2. A valid administrator account exists, and the administrator is already logged in.
3. A movie with a valid showtime already exists in the system.
4. The administrator has permission to manage showtimes.

1. The web browser is launched.
2. A valid administrator account exists, and the administrator is already logged in
3. The administrator has permission to manage movies.
4. The information for the new movie is available.

Test Steps
1. Open the customer login page. 2. Enter customer@test.com. 3. Enter ValidPass123!. 4. Click Log In.
1. Open the administrator MFA verification screen. 2. Enter 111111 as the MFA code. 3. Click Verify.
1. Generate a ticket for the selected movie, showtime, and Seat A10. 2. Record the generated ticket identifier. 3. Generate another ticket using separate valid test data. 4. Compare the two ticket identifiers and ticket details.
1. Create a transaction request for 21 tickets. 2. Submit the request to the transaction validation component. 3. Record the validation response.
1. Submit a valid payment request. 2. Receive an approved response from the payment service. 3. Open the customer purchase history. 4. Locate the completed transaction.
1. Search for the purchase using its transaction ID. 2. Submit the refund request. 3. Approve the refund through the payment service. 4. Retrieve the ticket and seat records.
1. Submit the checkout request to the payment service. 2. Approve the payment. 3. Verify that the ticket-generation service is called. 4. Retrieve the generated ticket.
1. Generate a ticket for Seat B5. 2. Retrieve the seat map for the same showtime. 3. Check the status of Seat B5. 4. Attempt to select Seat B5 for another transaction.
1. Navigate to the predefined URL. 2. Click the movie search bar located at the top-center of the page. 3. Enter "Interstellar" into the movie search bar. 4. Press Enter.
1. Select any smiley-face rating. 2. Click the Comment field and enter a comment. 3. Click the Submit button multiple times in quick succession.

1. Navigate to the predefined URL.
2. Select any available movie and showtime.
3. Select any available seat.
4. Select a ticket category.
5. Proceed to checkout without logging into a customer account.
6. Enter an invalid email address (aaa@gmail.com)
7. Click the Continue button

1. Navigate to the predefined URL.
2. Select any available movie, showtime, and theater location.
3. Select any available seat.
4. Select the Regular ticket category.
5. Proceed to checkout without logging into a customer account.
6. Enter a valid email address.
7. Click the Continue button.
8. View the Payment Summary section.

1. Launch the web browser
2. Navigate to the predefined URL
3. Click the Customer Login option
- 4.. Enter login credentials and Click Log in
5. Browse for a movie and select a movie, an available showtime
6. Select an available seat and proceed to Checkout
7. Review the ticket information and transaction total
8. Click Continue to Payment
9. Enter valid payment information and click Confirm Purchase
10. View the purchase confirmation and generated ticket.

1. Launch the web browser
2. Navigate to the predefined URL for administrator
3. Click the Administrator Login Option
4. Enter valid administrator login credentials and click log in.
5. Click the Refund Tickets option under ticket management section
6. Search for the completed purchase using the transaction ID.
7. Select the transaction and click process refund
8. Confirm the refund
9. view the refund confirmation and updated transaction details.

1. Click the Manage Showtimes section.
2. Click Modify Existing Showtime
3. Select any existing showtime to modify.
4. Click Edit Showtime
5. Modify the showtime date and time.
6. Click Save Changes.
7. Return to the administrator home page and search for the updated showtime by entering the movie title in the search bar at the top of the page
8. Click the movie and locate the modified showtime
9. View the updated showtime details.

1. Click the Manage Movies section.
2. Click Add New Movie
3. Fill out the Movie Information form includeing the title, genre, duration, and description
- 4, Enable Show IMDb rating and Show Rottentomatoes rating
5. Fill out the Auditorium number and Showtime sections.
6. Click Save and Publish
7. Return to the Administrator home page
8. Click the search bar at the top and type the newly added movie title
9. Click the newly added movie and view its details

Expected Result	Actual Result	Status
The credentials should be validated, an authenticated customer session should be created, and the customer account page should be displayed.	The credentials are validated, an authenticated customer session is created, and the customer account page is displayed.	Pass
The MFA verification should fail, administrator access should be denied, and a clear error message should be displayed.	The MFA verification fails, administrator access is denied, and a clear error message is displayed.	Pass
Each ticket should contain the correct movie, showtime, and seat information, and each generated ticket identifier should be unique.	Each ticket contains the correct movie, showtime, and seat information, and the generated ticket identifiers are unique.	Pass
The transaction should be rejected and an error message should state that no more than 20 tickets may be purchased in one transaction.	The transaction is rejected and an error message states that no more than 20 tickets may be purchased in one transaction.	Pass
The payment should be approved, a completed transaction record should be created, and the transaction should appear in the customer purchase history with the correct amount and ticket information.	The payment is approved, a completed transaction record is created, and the transaction appears in the customer purchase history with the correct amount and ticket information.	Pass
The refund should be completed, the ticket status should become Refunded, the digital ticket should be invalidated, and the seat should become available.	The refund is completed, the ticket status becomes Refunded, the digital ticket is invalidated, and the seat becomes available.	Pass
The system should generate one unique ticket for each purchased seat only after payment approval, with the correct transaction, movie, showtime, and seat information.	The system generates one unique ticket for each purchased seat only after payment approval, with the correct transaction, movie, showtime, and seat information.	Pass
Seat B5 should be marked as sold or unavailable immediately after ticket generation, and another customer should not be able to select it.	Seat B5 is marked as sold immediately after ticket generation, and another customer cannot select it.	Pass
The search results related to "Interstellar" should be displayed.	The search results related to "Interstellar" are displayed	Pass
Only one feedback submission should be processed and recorded in the Theater Ticketing System database. Duplicate feedback submissions should be prevented.	Only one feedback submission is processed and recorded in the Theater Ticketing System database. Duplicate feedback submissions are prevented.	Pass

A CSS styled error message indicating that the email address is invalid should be displayed after the Continue button is clicked. The user should remain on the guest checkout page and the system should not proceed to the next page.	A CSS styled error message indicating that the email address is invalid is displayed after the Continue button is clicked. The user remains on the guest checkout page and the system does not proceed to the next page.	Pass
The 0% tax, a 5% online service fee, and a \$2.99 seat reservation fee should be added to the ticket price when calculating the total ticket price. The Payment Summary section should display the correct total ticket price.	The 0% tax, a 5% online service fee, and a \$2.99 seat reservation fee are added to the ticket price when calculating the total ticket price. The Payment Summary section displays the correct total ticket price.	Pass
The homepage should be displayed correctly. The user should be successfully authenticated. Available movies, showtimes, and seat should be displayed. The selected seat should be added to the checkout. The transaction amount should be displayed correctly. The purchase should be completed successfully. The purchased seat should be marked as unavailable for future purchases. A valid ticket should be generated with the correct movie, showtime, seat number and transaction details. A purchase confirmation page should be displayed to the customer.	The homepage is displayed correctly. The user is authenticated and available movies, showtimes and seats are displayed and selectable. The transaction amount is correct, and the ticket purchase is completed successfully. The purchased seat is unavailable for future purchases. A valid ticket and purchase confirmation page are displayed.	Pass
The administrator should be authenticated successfully. The Refund option should be displayed. The completed purchase should be found using the transaction ID. The refund should be processed successfully. A refund confirmation page and transaction details should be displayed correctly. The status of the ticket should be marked as "refunded" and the refunded seat should be available for future purchases.	The administrator is authenticated successfully. The Refund Tickets option is available and the completed purchase is found using the transaction ID. The refund is completed successfully. A refund confirmation page and updated transaction details are displayed correctly. The ticket status is marked as "refunded" and the refunded seat is available for future purchases.	Pass

The Manage Showtimes, and Modify Existing Showtime sections should be displayed and available to the administrator. The selected showtime should be available for modification. The modified showtime information should be saved successfully. A confirmation message should be displayed in the top-right corner of the page. The updated showtime details should be the same as the modification made by the administrator and displayed correctly.	The Manage Showtimes and Modify Existing Showtime options are displayed and available. The selected showtime is available for modification, and the modified showtime is saved successfully. A confirmation message is displayed in the top-right corner of the page. The updated showtime details match the modifications made by the administrator.	Pass
The Manage Movies and Add New Movie options should be displayed and available to the administrator. The movie information should be accepted. The movie should be saved and published to the public successfully. A confirmation message should be displayed in the top-right corner of the page. The newly added movie information should match the information entered by the administrator and be displayed correctly.	The Manage Movies and Add New Movie options are available. The movie information is accepted, and the new movie is published successfully. A confirmation message is displayed in the top-right corner of the page. The newly added movie information is matches the information entered by the administrator and is displayed correctly.	Pass

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